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The impact of a chatbot working as an assistant in a course for supporting student learning and engagement

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Abstract

Nowadays, many digital learning tools can be integrated into a course to enhance the teaching–learning process. It is a fact that technology is changing how students learn and how professors teach. The growing utilization of AI chatbots in education to support students' learning underscores the importance of identifying critical aspects related to students' acceptance of this technology. Teachers require a deeper understanding of how students perceive and respond to chatbot assistance, as well as methods to measure levels of satisfaction, engagement, and knowledge acquired while using these technological tools. The aim of this study was, on the one hand, to illustrate the integration of a chatbot into a course to assist students in solving specific learning activities. On the other hand, it aimed to analyze the correlation between the factors influencing university students' intention to use chatbots and their perceived usefulness. It is noteworthy that the learning activity integrated two technologies: an Integrated Development Environment and a chatbot, offering additional educational value to students. The Technology Acceptance Model was used to measure the students' perceived usefulness and attitude toward this digital learning tool. The findings of this study provided insights to faculty on how to integrate and use chatbots to enhance students' engagement and motivation. The results revealed that the perceived usefulness of chatbots does, in fact, influence students' intention to use them. Most of the students expressed satisfaction with having a chatbot to assist them in solving exercises. Students highlighted that one of the most significant benefits was the 24/7 chatbot's support availability and its user-friendly interface.

KEYWORDS

chatbots, digital learning tools, engagement, teaching and learning strategies

1 | INTRODUCTION

Nowadays, many digital learning tools can be integrated into a course to enhance the teaching–learning process. It is a fact that technology is changing how students learn and how professors teach. Digital learning tools can help students learn faster, improve students´ engagement with subjects, and increase the effectiveness of teaching [52].

Students appreciate how these tools offer them guidance, positive feedback, and enjoyable digital learning experiences. In general terms, these tools promote active, collaborative, creative, and evaluative learning experiences. Digital learning tools can provide learners with insights into their progress and bring new possibilities for formative feedback, skill development, and self-reflection [45].

Chatbots, also known as conversational pedagogical agents in the education field, and Integrated Development Environments (IDEs), are educational technologies that facilitate distance learning, enabling students to progress at their own pace while accessing learning materials ubiquitously [60]. Several studies indicate that integrating these tools into course activities supports students' learning processes, enhances their effectiveness, and plays a crucial role in promoting student engagement [10], transforming students into active learners. The integration of these technologies improves students' chances of successfully completing tasks or entire courses [40].

During the COVID-19 pandemic, the adoption of diverse digital tools became imperative, aiming to not only enhance students' commitment and motivation but also to offer a viable alternative to traditional, in-person activities conducted during classes or laboratory sessions. This unprecedented situation compelled both teachers and students to embrace the digital academic experience [33].

A chatbot is an Artificial Intelligence (AI) program designed to simulate human conversations naturally, either through voice interfaces or text messages [21]. Chatbots are typically integrated into applications, websites, or instant messaging platforms. In the education field, chatbots can be employed in various capacities, serving as virtual teacher assistants to help students solve course assignments or learn concepts of a subject. Chatbots serve a variety of functions, such as asking questions, offering responses, retrieving information, and providing valuable insights. Based on the premise that these are the actions that a good teacher-assistant would provide to a student (personalized support to improve students' performance), chatbots can be utilized to establish the knowledge that students need to learn and provide them with a roadmap, answers, or tips [43]. An IDE is a computer application designed to integrate all programming tasks into one application and facilitate the development of other applications. It offers a centralized interface with all the tools a developer or programmer needs [13]. Codeboard is an IDE developed specifically for the Java language, providing suggestions and guidelines for users to edit and write Java code [58]. In programming courses, theory and practice cannot be separated; if the course focuses solely on theoretical aspects, there is a risk of not achieving the learning outcomes. Furthermore, theory without practice would be insufficient, and students would not engage in the stages of analysis, practice, and reflection that occur in a natural learning cycle [36]. The most significant benefit of using an IDE is that students can code and run Java programs on their own computers."

The growing utilization of AI chatbots in education to support students' learning underscores the importance of identifying critical aspects related to students' acceptance of this technology. Teachers require a deeper understanding of how students perceive and respond to chatbot assistance, as well as methods to measure levels of satisfaction, engagement, and knowledge acquired through these technological tools.

In this study, the chatbot serves as an instructional medium to enhance learning. However, to efficiently utilize it in a course, in addition to considering cognitive factors such as memory, attention, problem-solving skills, and information processing, we must gain a better understanding of student acceptance and the factors influencing their adoption and usage. Therefore, it is necessary to investigate students' perceptions and behaviors related to the use of chatbots in their assignments.

With the rise of generative AI tools, such as ChatGPT or Bard, this study can provide educators with information on how to integrate conversational agents into their courses and what factors must be considered. The aim of this study was, on the one hand, to illustrate the integration of a chatbot into a course to assist students in solving specific learning activities. On the other hand, it aimed to analyze the correlation between the factors influencing university students' chatbot intention to use and perceived usefulness. The findings of this study hold the potential to contribute valuable insights toward optimizing chatbot integration in courses, enhancing student learning, and fostering a deeper understanding of the factors shaping students' acceptance and utilization of this innovative educational tool.

2 | BACKGROUND

The implementation of chatbots is not very extensive in educational environments compared to traditional intelligent tutoring systems [22]. Several factors may explain this disparity. First, it is essential to consider the inherent complexity of educational processes and the diversity of student needs. Although chatbots offer the advantage of conversational interaction, precise adaptation to pedagogical requirements remains an ongoing challenge; in contrast, intelligent tutoring systems have often been developed over decades, enabling them to provide a deeper and proven level of personalization and adaptability [57]. Furthermore, the time and resource investment required to develop and maintain high-quality educational chatbots has deterred many institutions. An effective educational chatbot must be supported by extensive databases, advanced language models, and regular updates to reflect changes in the curriculum and pedagogy [32].

On the other hand, established intelligent tutoring systems have a solid foundation and a proven track record [7], which may make institutions hesitant to switch. However, it is important to emphasize that chatbots offer unique advantages in terms of accessibility, flexibility, and immediate feedback that cannot be matched by traditional intelligent tutoring systems. A recent study [35] highlighted how the continuous feedback provided by chatbots can significantly enhance individualized learning.

While it is true that the implementation of chatbots in educational environments is not as extensive as that of traditional intelligent tutoring systems, it is important to recognize that chatbots have great potential to improve education, significantly impacting learning success and students' satisfaction [19].

Georgia University developed a chatbot called "Jill Watson," which was developed to manage forum posts by students enrolled in a programming course. The development team used IBM's Watson Platform. The chatbot was initially tested in a pilot study in 2016, where it successfully answered student questions and provided useful feedback. After the pilot study, the chatbot was refined and improved, and it was integrated into the online course. Alongside a team of human teaching assistants, Jill answered student questions as they came in. At the end of her semester-long debut, students were not able to distinguish which of the TAs was the AI.

Since its launch, Jill Watson has been used in several courses and has answered thousands of student questions. The chatbot has been well-received by students and has helped to improve their learning experience. Overall, Jill Watson is an excellent example of how AI-powered chatbots can be used to enhance education and assist learners in their studies. The study results showed that students were more engaged, and they expressed they would like to have this type of support in other courses [8].

Quiroga Pérez [46] and Dina [3] are chatbot applications used to mitigate the overload of student questions during enrollment periods by the University of Murcia in Spain, and Dian Nuswantoro Semarang University, respectively. The chatbots were developed to assist students with enrollment-related queries such as course schedules, availability, and registration procedures. Both use natural language processing (NLP) to understand and respond to student questions. They were trained on a large data set of enrollment-related questions and answers, which allowed them to provide accurate and relevant responses to student queries [3].

The chatbots were launched in 2018 and have been well-received by students, who appreciate the convenience of being able to quickly access information and receive answers to their enrollment-related questions. The chatbots have also helped to reduce the workload of university staff by automating the handling of routine student inquiries. Lola and Dina are excellent examples of how chatbots can be used to streamline administrative processes and enhance the student experience [1].

Duolingo and Ani serve as examples of chatbots utilized in informal education. Duolingo, a language-learning platform offering courses in numerous languages, employs gamification techniques to enhance the fun and engagement of language learning. The platform utilizes interactive exercises, quizzes, and personalized feedback to aid students in practicing listening, speaking, reading, and writing skills in the target language [28]. On the other hand, Ani, developed by Duolingo, works as a chatbot designed to assist learners in practicing conversation skills in a foreign language. Ani employs NLP to comprehend and respond to user queries in the target language. It provides personalized feedback to enhance pronunciation and grammar and can suggest conversation topics based on user interests and preferences [20].

Both Duolingo and Ani are valuable tools for language learners as they adapt to each user's learning pace and style. Duolingo is used by millions of learners globally, and it has garnered positive reviews for its effectiveness and user-friendly interface. Additionally, some chatbot applications incorporate gamification techniques to sustain learner motivation. By integrating game-like elements into the learning process, these applications keep learners engaged, motivated, and more likely to achieve their learning goals, as seen in the example of Mobile Chatbot [44]. Chatbots play a crucial role in helping students analyze problems and receive timely feedback. These benefits are highly valuable for students and drive their engagement with digital learning tools.

Chatbots are computer programs that offer a conversational experience by leveraging AI and, specifically, NLP, a subfield of AI. They aim to mimic an intelligent human conversation with real people [24]. A chatbot is a term that is a mix of two words: converse (chat) and robot (bot). When they attend to the specific needs of users, they are known as "virtual agents" or "personal agents" [54]. Some chatbots are built using Artificial Intelligence Markup Language (AIML), a language developed to function as the brain of chatbots, such as the one created by Dr. Wallace at the Alice Foundation [38]. However, it is worth noting that chatbots may also utilize alternative technologies and programming languages.

Original chatbots recognize patterns and rules, but the advanced ones utilize Deep Learning. In the context of NLP, deep learning can be described as recognizing patterns and even rules in text embeddings. User experience (UX) and User Interface (UI) [30] are also vital in chatbots as they enable establishing a natural, intelligent, and coherent conversation. Some advantages of chatbots include their ability to learn from past interactions and improve responses over time. Users can also efficiently perform tasks autonomously.

In education, chatbots operate on the premise that teaching is an act of communication and interaction. Chatbots can address repetitive questions with straightforward solutions or assist students in finding the correct answers to simple assignments. However, it is crucial to emphasize that teachers remain the backbone of the teaching process. Technology, including chatbots, should be employed as an assistant rather than a replacement [51].

Chatbots are being introduced everywhere, for example, as customer service [63] or as information services [39]. Unfortunately, not many chatbot applications can be found in traditional courses. The reason for this could be that the capabilities and potential usages of chatbots in education are still unknown, and there is little literature about it. However, a type of chatbot has been developed to foster teaching and learning, known as 'chatbots with educational intentionality. They can be categorized into two groups: (a) Tutors, where the main task of the chatbot is to adapt, select, and sequence content to provide learning motivation according to the student's needs and pace; and (b) Exercise assistants, where the chatbot presents a question or problem, and the student answers it. Subsequently, the chatbot automatically assesses the student's answer and provides immediate feedback. The latter category is utilized for the acquisition of skills [31]. Recent research studies indicate that skill improvement is the most popular objective, followed by the efficiency of education, with student motivation ranking third [62].

Chatbots, with well-designed instructional content, can be integrated into a course, enabling students to explore various subjects, engage in learning activities, and receive assistance from the chatbot throughout the process. As the integration of new technologies becomes increasingly prominent in education, it is essential to consider the acceptance or rejection of these technologies by potential users, such as students and teachers [15]. This concern underscores the significance of frameworks like the Technology Acceptance Model (TAM), which provides insights into the factors influencing the adoption and utilization of technology in educational settings [12, 50].

2.1 | The TAM

The TAM, rooted in the Theory of Reasoned Action (TRA) and the Theory of Planned Behavior [5], aims to understand how end-users decide to adopt new technologies. TAM's popularity stems from its broad applicability to diverse technologies and users [59]. It explains user motivation based on perceived usefulness, perceived ease of use, and attitude toward using a system [34].

Perceived usefulness measures the belief that using a system improves task performance, and perceived ease of use gauges the belief that using a system requires minimal mental and physical effort [14]. Both constructs are directly influenced by system features (Figure 1). As technology evolves, new factors continuously emerge, influencing the TAM model's core. Researchers have adapted TAM, leading to studies labeled TAM++, where additional constructs are introduced [9, 17]. Figure 1 depicts the original TAM model by Davis.

TAM's validity is well-established, with applications in various fields, including healthcare, e-business adoption, social media, wireless Internet, e-shopping, and education [2, 11, 29, 37, 53, 61]. In education, TAM has been used to predict technology acceptance by teachers and students across different domains, such as Personal

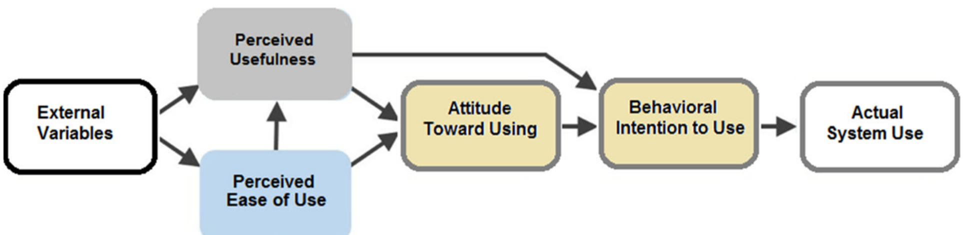

FIGURE 1 Technology Acceptance Model (TAM) from Davis, 1989.

Learning Environments (PLEs) and mobile learning [18, 49]. It remains a recognized framework for assessing learning technology use [23].

The TAM model explores factors influencing individuals' acceptance, rejection, or continued use of technologies [4]. It provides a theoretical context explaining the relationship between attitudes, intentions, and behavior. User attitudes are shaped by perceived usefulness and perceived ease of use, with both factors directly impacting technology use [48]. Behavioral intention to use and actual use are outcome variables, with the model applicable in various contexts, leading to the development of extended models incorporating external variables [27]."

3 | INTEGRATING A CHATBOT IN A COURSE: EXPERIMENTAL DESIGN

In this section, we delve into the experimental design employed to integrate a chatbot into a first-year course in the Computer Engineering program at Galileo University. We outline the methodology and key considerations that guided the experimental design. The experiment involves developing a learning activity utilizing a chatbot and an Integrated Development Environment (IDE) to solve a Java programming problem.

The aim of the course, "Evolution of Science and Technology," is to delve into diverse technological phenomena that have ignited notable inventions and advancements, shaping the trajectory of human evolution. In this context, the incorporation of a learning activity that employs a chatbot as an illustration of emerging technology was deemed a highly beneficial initiative. Annually, around 200 students enroll in this course, divided into six groups for instructional purposes. The course spans 18 weeks, each lesson has video lectures, learning activities, and a questionnaire. Academic support is extended to students through various channels, including tutoring sessions, lectures, forums, and email.

The learning activity was named "Fix it (Hands-on work)." Its main objectives were to help students reinforce the concepts learned about the Java language and to provide them with the opportunity to practice at their own pace, as many times as needed, with the assistance of the chatbot. As part of the activity, students received a Java program code containing errors, and their task was to identify and correct these errors with the assistance of a chatbot and Codeboard.

The errors in the given Java basic program were classified into four categories. The chatbot displayed a menu with these categories, and according to the instructions given by the chatbot, students had to select options "a," "b," "c," or "d" to find and correct the error corresponding to each category. Following the idea of imitating an intelligent human conversation, the AI chatbot asked questions to students, and vice versa. With each interaction, students received new information and instructions that allowed them to fix the error. Subsequently, they had to compile and run the corrected Java program using Codeboard to ensure the solution was accurate. If the error persisted, students had to repeat this process until they found the correct answer with the chatbot's help. Students had to be very careful with their corrections to the Java program code because the changes could cause infinite loops. Figure 2 illustrates the process.

The design of the learning activity was carried out in three phases. The first one consisted of collaborating with the course teacher to determine the main learning topics and understand what students should be able to do by the end of the assignment. The deliverable of this phase was a comprehensive description of the activity, instructional outcomes, instructions to students, and the main role of the chatbot, acting as an assistant.

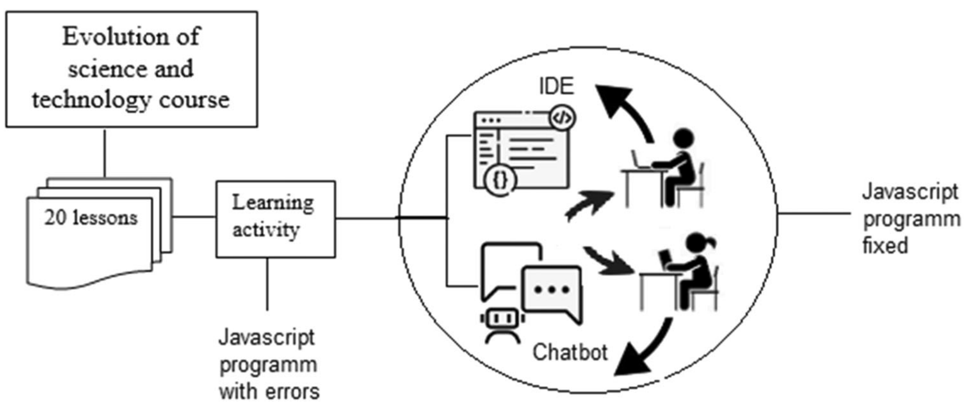

FIGURE 2 Learning activity process.

In the second phase, the type of chatbot was selected, along with determining how to embed it into the learning scenario. A flowchart chatbot was designed based on a predefined flowchart or decision tree structure. In this type of chatbot, the conversation with the user follows a specific path, and the chatbot's responses are determined by a set of predefined rules or conditions. This structured approach allows the chatbot to guide users through a series of questions or choices, leading to specific outcomes or responses based on the user's inputs [41]. This method not only allows students to interact and move forward with the conversation, but it also allows the chatbot to save students' responses for future analytics.

A rule-based chatbot, by default, does not possess the capability to learn or comprehend human text/voice input, meaning it does not incorporate AI unless the underlying code is specifically modified. In this instance, the source code was adapted to integrate AI features. During this phase, we obtained precise data within the critical domain of the course subject to train the chatbot, enabling it to effectively assist and guide students in their problem-solving activities. The effectiveness of a chatbot fundamentally depends on the quality of its meticulously curated training data. In this context, the data was sourced from the teacher's expertise and insights, as well as contributions from some of the top-performing students. The third stage involved constructing and integrating the chatbot into the LMS. The chatbot's UI should prioritize ease of use, intuitive navigation, and visual appeal to ensure an enjoyable user experience. Key features include a scrollable chat history, quick replies to suggestions, clear communication, and seamless integration with the platform. By incorporating these elements into our chatbot's development, we aimed to create a user-friendly interface that enhances students' learning experiences.

4 | CHATBOT ARCHITECTURE AND CHALLENGES

4.1 | Chatbot architecture

The development team explored various platforms as alternatives to implement a chatbot. They specifically sought solutions that aligned with the learning activity's objectives and chose Google Dialogflow combined with machine learning. Google Dialogflow relies on Natural Language Understanding (NLU), a key aspect of Natural Language Processing (NLP), which deals with the crucial task of managing unstructured inputs and converting them into a structured format that machines can interpret and act upon [55]. While Google Dialogflow provides a user-friendly interface and pre-built templates to simplify chatbot development, some level of programming knowledge may still be necessary for more advanced customization and integration tasks. Nevertheless, Dialogflow significantly reduces technical complexity and coding requirements. With its intuitive interface and built-in natural language processing capabilities, Dialogflow enables users to create functional chatbots without extensive programming expertise.

To create the chatbot, five main components were used: (a) Dialogflow, an NLU platform used to analyze text or audio inputs that allow programmers to integrate a conversational UI into the chatbot. (b) Cloud Firestore, which is a flexible and scalable NoSQL database. It stores and synchronizes data for the client and server side. The implementation was carried out by using a Cloud Function that would receive and parse the request payload from Dialogflow and update the Firestore database [42]. (c) Cloud Functions for Firebase allowed the development team to automatically run backend code in response to HTTPS requests and events triggered by Firebase features [64]. (d) Rocketbots is a cloud-based chat automation platform designed to help managers collect client information and segment customers. It integrates with various messaging social network channels, such as Facebook, Workspade, Slack, Kik, and many others. Rocketbots acts as a bridge between managers and users. Rocketbots was used for monitoring and tracking students' interactions,by identifying frequently asked questions and assessing the overall performance of the chatbot. (e) LMS In this architecture, the learning management system (LMS) used was the Galileo Educational System (GES), which is an in-house built system. The chatbot was embedded into the LMS via inline frames with the exchange of communication. Figure 3 shows the chatbot architecture.

NLU is a subset of AI that leverages a combination of statistical, predictive analytics, data modeling, and data mining techniques to extract meaningful information. It is crucial to acknowledge that the effectiveness of a chatbot is intricately tied to the quality of its training data. Our data set is curated from the teacher's personal knowledge and insights, as well as contributions from top-performing students.

To analyze text input, NLU employs a hierarchy of classification models. Initially, it categorizes input into predetermined conversational domains, such as "Java programming." Subsequently, it identifies the student's intent by assigning each input to predefined intents in the NLP algorithm. The entity recognizer then extracts essential phrases and words necessary to fulfill the student's intent. An "entity" contains knowledge and behaviors used to respond to phrases and meet user intentions.

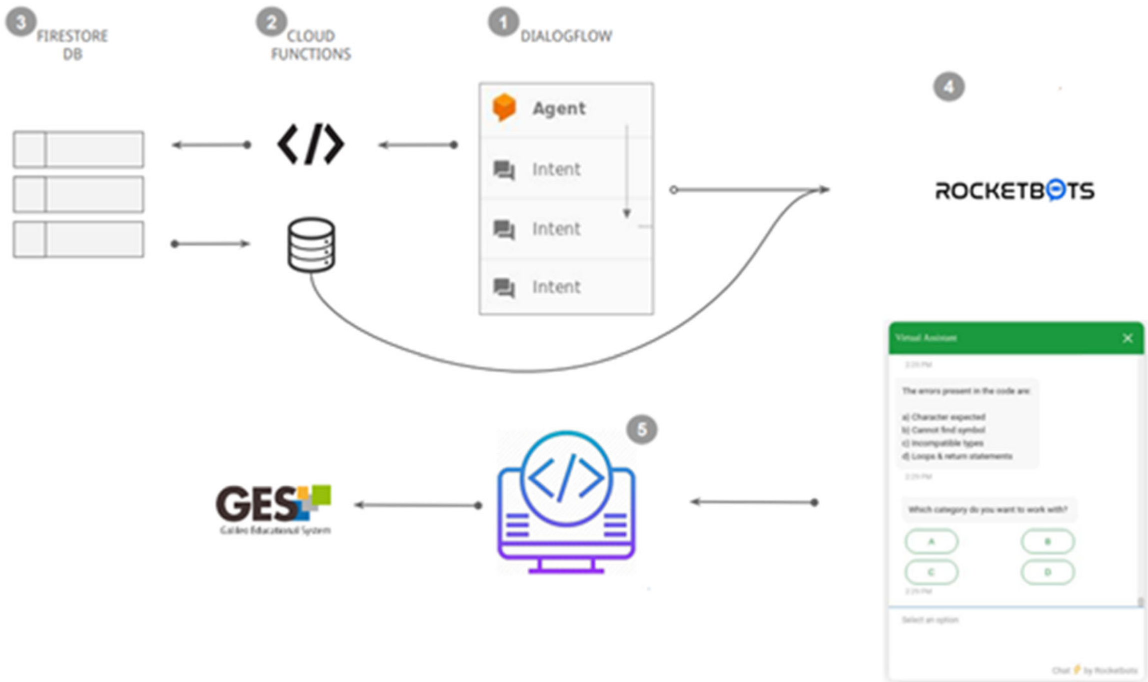

FIGURE 3 Architecture components of the AI Chatbot Assistant.

In our approach, the chatbot was trained using a retrieval-based method, selecting the best response from a predefined database [66]. These responses were carefully crafted based on information provided by the teacher and students. Additionally, the chatbot engages students by posing follow-up questions through a defined template and generates the rest of the sequence by considering student inputs and context. This approach ensures that responses are tailored to individual needs.

Importantly, responses are generated through machine learning, specifically utilizing generative models [65]. This means that the chatbot's answers are not only based on predefined responses but also on the patterns and structures it has learned from the training data, enhancing its ability to provide contextually relevant information.

Figure 4 illustrates the learning activity implemented into the LMS.

The level of students' engagement and motivation was measured by their satisfaction and perceived benefits from interacting with the chatbot. In the context of our research, satisfaction pertains to students' contentment with the speed, accessibility, and effectiveness of using chatbots as an educational tool.

4.2 | Challenges

When incorporating a learning activity of this nature into a course, two distinct challenges arise. The first centers on crafting the activity, demanding explicit instructional objectives and profound subject matter expertise from the teacher. Collaborating with an instructional designer becomes imperative to ensure the activity is meticulously structured and aligns seamlessly with student needs. The second challenge pertains to programming the chatbot, presenting various hurdles for programmers. These encompass critical decisions such as platform selection, addressing NLP intricacies, optimizing Cloud Functions, and seamlessly integrating tools like Rocketbots. Essentially, these challenges span both the instructional design phase and the technical implementation aspects, each necessitating distinct expertise and careful consideration.

5 | RESEARCH MODEL

As mentioned earlier, the present research was conducted in a first-year course for engineering students. The model employed is rooted in the TAM. The research question guiding this study was: “To what extent are first-year engineering students willing to embrace and utilize chatbots as a supportive tool? Additionally, what are the primary factors influencing their perceived ease of use and perceived usefulness, as outlined by the Technology Acceptance Model?”

This question is examined in terms of four specific hypotheses that are directly based on TAM:

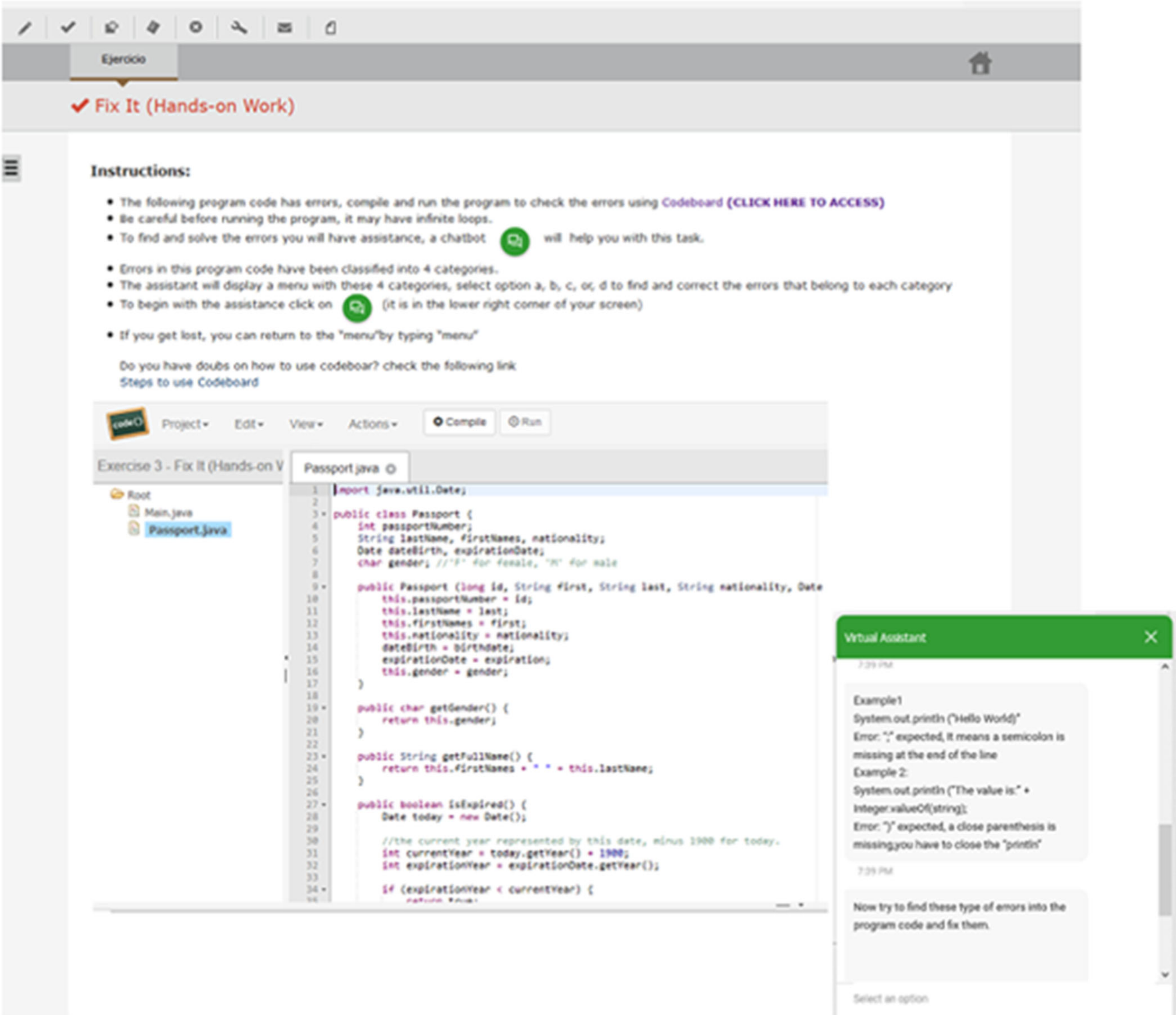

FIGURE 4 Learning Activity implemented into the LMS (GES).

- H1: Perceived ease of use positively affects students' perceived usefulness towards chatbots.
- H2: Perceived usefulness has a positive influence on students' intention to use chatbots.
- H3: Perceived usefulness has a positive influence on students' attitude toward chatbots.
- H4: Attitude has a significant positive effect on student's intention to use chatbots.

Building upon the construct principles of TAM, we constructed a research model to elucidate the research question through the lens of the hypotheses. The analysis of the model will allow us not only to analyze and understand the relationships between the constructs but also to measure their reliability and validity, thereby furnishing valuable insights into the factors impacting the adoption and utilization of chatbots. This information can be useful in crafting interventions to enhance students' acceptance of chatbots as a supportive tool in tackling Java programming problems. Figure 5 illustrates the research model.

The proposed research model uses four constructs. Table 1 shows the definition and abbreviation of each one.

6 | METHOD

6.1 | Participants

Two hundred students were enrolled in the course, but only 73 students actively participated in the research. It is

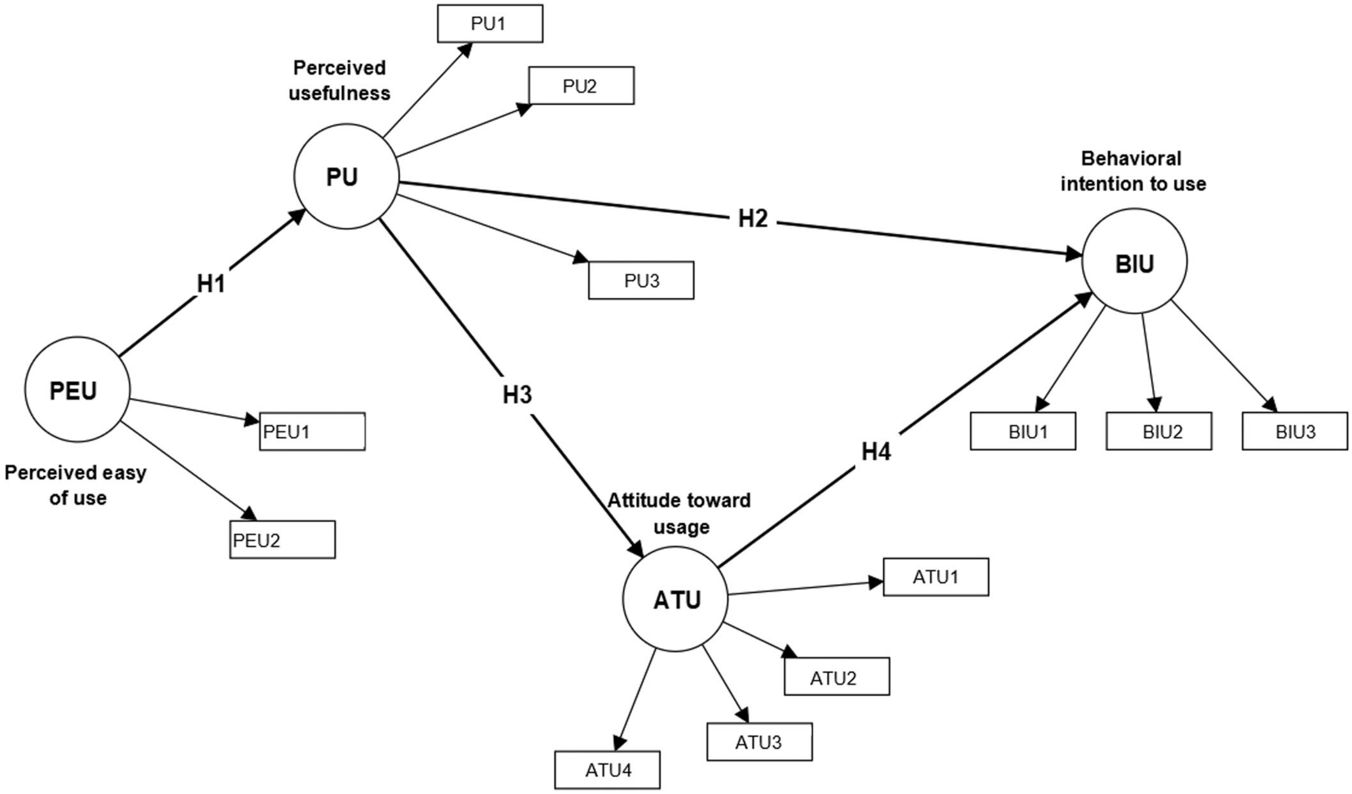

FIGURE 5 Research model.

important to note that it was not mandatory to answer the survey at the end of the activity. Seventy-two percent of the participants considered themselves intermediate-level technology users. Fifty-one percent of the participants said that they already knew about the usage of chatbots in learning contexts. Sixty of the respondents were male, and 13 were female. Eighty-five percent of the respondents were below the age of 20, and 15% of respondents were between 20 and 29 years old. There was some concern about the gender ratio; however, it is common in this type of course. Descriptive statistics related to the respondents' demographic details are shown in Table 2.

6.2 | Data collection

A 5-point Likert-type scale was employed to collect the data, where 1 represented "strongly disagree," 2 meant "disagree," 3 represented "neutral," 4 was "agree," and 5 represented "strongly agree." The first section of the questionnaire gathered participants' general demographic information and technology usage level. The second section assessed perceived usefulness and perceived ease of use, while the third section measured attitude toward usage, level of satisfaction, and behavioral intention to use. Measurements for PU, PEU, ATU, and BIU were adapted from Davis' [16] study. The items for each construct are described in the Appendix.

In addition to the instrument, each participant was provided with an information sheet outlining the nature and purpose of the research, along with a consent form to be completed prior to the survey.

6.3 | Data analysis

Due to the small sample size ($N = 73$) of the survey, structural equation modeling (SEM) using partial least squares (PLS) regression was employed. The SmartPLS 4 program was utilized to assess the research model and examine the hypotheses. The reliability and validity of our research model were assessed using the outcomes of the factor analysis, ensuring the robustness of our constructs. To evaluate the structural model's fit, *P*-tests and bootstrapping were employed.

To assess the reliability of constructs, Cronbach's α analysis and composite reliability (CR) values, both with a limit of 0.7, were employed to validate internal consistency. Table 3 demonstrates that the constructs meet both conditions, indicating that the scales are

TABLE 1 Construct definition.

Abbreviation	Construct	Definition
PU	Perceived usefulness	The degree to which the students think that using the chatbot will enhance their academic performance.
PEU	Perceived ease of use	The degree to which the students think that using the chatbot is effortless.
ATU	Attitude towards usage	It reflects the students' overall perception regarding the use of chatbots in their coursework. It represents whether they hold a favorable or unfavorable attitude towards employing chatbots as an educational tool, which can significantly influence students' acceptance.
BIU	Behavioral intention to use	It represents the students' willingness and intention to incorporate chatbots into their coursework. It serves as a strong predictor of their actual use of chatbots for course assignments.

TABLE 2 Descriptive statistics of respondents' demographic details.

Measure	Value	Frequency	Percent
Gender	Male	60	82
	Female	13	18
Age	Below-20	62	85
	20–29	11	15
Technology Usage level	Advanced user	10	14
	Intermediate user	10	72
	Novice user	53	14
Knowledge of chatbots	Yes	37	51
	No	36	49

reliable. Convergent validity was evaluated using the average variance extracted (AVE) based on the 0.5 threshold. The AVE value for each construct exceeds 0.5, satisfying the criterion for convergent validity, as presented in Table 3.

To assess discriminant validity, an examination of the correlation matrix alongside the AVE values was carried out. As outlined in Table 4, the AVE values for all measures consistently exceeded the off-diagonal correlations. Therefore, the discriminant validity among the variables, in accordance with Fornell and Larcker's criterion (1981), is deemed satisfactory.

A global fit measure (GoF or Goodness of Fit) for PLS path modeling was conducted, defined as the geometric mean of the average communality and average R2 for endogenous constructs [56]. We obtained a GoF value of 0.6463, surpassing the cut-off value of 0.36 [6]. Additionally, we obtained an SRMR value of 0.072, suggesting a relatively small discrepancy between the observed and model-implied covariance matrices, supporting the adequacy of the proposed structural model. This indicates that the model provides a reasonable representation of the relationships among the observed variables, validating its fit to the data.

7 | RESULTS

The model derived from the PLS path analysis is shown in Figure 6. This figure illustrates the path coefficients along with their associated significance tests (as determined through bootstrapping) and the multiple determination coefficients.

The results of the hypothesis testing for H1 (0.877; $p < .001$) strongly support the hypothesis that “Perceived ease of use positively affects students' perceived usefulness towards chatbots.” This implies that students are more likely to perceive chatbots as valuable and beneficial in their educational context when they find them easy to use.

The results of hypothesis testing for H2 (0.290; $p < .001$) are statistically significant, providing strong evidence in favor of the hypothesis that “Perceived usefulness has a positive influence on students' intention to use chatbots.” This outcome indicates that students' perceptions of how the chatbot enhances their academic performance effortlessly using chatbots have a substantial and positive influence on their overall attitude toward using chatbots as a tool in their course assignments.

The results of the hypothesis testing for H3 (0.699; $p < .001$) provide strong support for the hypothesis that “Perceived usefulness has a positive influence on students' attitude towards chatbots.” This indicates that students are more likely to hold a positive attitude toward chatbots when they perceive them as useful in their educational experience. The statistically significant result underscores the importance of perceived usefulness in shaping students' attitudes, suggesting that students value chatbots more positively when they perceive them as beneficial in their academic pursuits.

TABLE 3 Constructs and indicators.

Constructs	α	Items	Factor loadings	CR (rho_a)	CR (rho_c)	AVE
ATU	0.776	ATU1	0.741	0.786	0.856	0.598
		ATU2	0.718			
		ATU3	0.807			
		ATU4	0.823			
BIU	0.791	BIU1	0.785	0.822	0.875	0.701
		BIU2	0.851			
		BIU3	0.877			
PEU	0.958	PEU1	0.982	0.967	0.979	0.960
		PEU2	0.977			
PU	0.797	PU1	0.790	0.831	0.88	0.711
		PU2	0.831			
		PU3	0.904			

TABLE 4 Construct correlations and the square root of average variance extracted values.

	Attitude toward usage	Behavioral intention to use	Perceived ease of use	Perceived usefulness
Attitude toward usage	0.773			
Behavioral intention to use	0.703	0.838		
Perceived usefulness	0.609	0.611	0.843	
Perceived ease of use	0.522	0.571	0.877	0.980

The results of H4 ("Attitude has a significant positive effect on student´s intention to use chatbots") were 0.526 ($p < .001$). The significant positive coefficient of 0.526 suggests that when students perceive chatbots positively and perceive them as beneficial for their learning, they are more inclined toward actively adopting and using chatbots in their course assignments. Table 5 shows the relationships.

Table 6 presents the primary benefits identified by students when utilizing a chatbot as an assistant to solve exercises.

To collect this data, the assistant teacher asked students about the benefits they found while using the chatbot. The most frequently mentioned benefits were then listed in a question within the questionnaire. Overall, using a chatbot can offer various advantages to students, including faster response times, personalization, and 24/7 availability. Notably, 35.6% of students consider the main benefit of a chatbot to be its ability to easily detect errors, while 27.4% believe that a chatbot can provide valuable support and enhance the learning experience.

8 | DISCUSSION

This study presented a specific application of chatbots in education. Despite the small data set, the results were interesting, demonstrating how a chatbot could be effectively employed for educational purposes, acting as a tutor to assist students in their learning process.

The findings revealed a significant positive relationship between students' perceptions of the usefulness of chatbots and their overall attitude toward utilizing them. This underscores the potential of incorporating chatbots as educational tools, suggesting that when students recognize the practical benefits and effectiveness of chatbots in addressing their queries and aiding in the learning journey, their attitudes become more favorable. The mentioned conclusion aligns with the findings from the case of "Jill Watson" at Georgia University [7], where chatbots are being designed and implemented to enhance various aspects of the learning process. Our study was grounded in the premise that chatbots when implemented effectively, can enhance the learning experience in several ways. The ability of the chatbot to provide quick

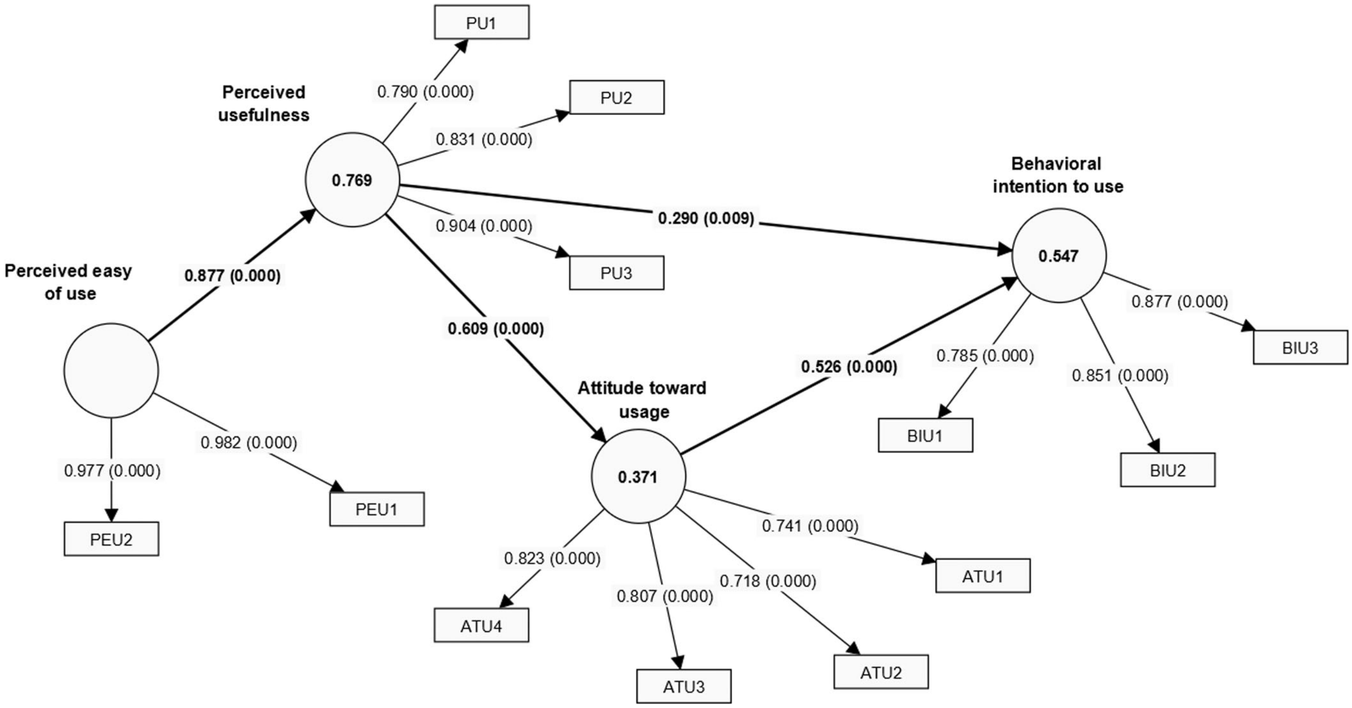

FIGURE 6 The results of partial least squares regression.

TABLE 5 Relationships, path coefficients—mean, SD, and *p*-values.

	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	*p* Values
Attitude toward _usage -> Behavioral _intention to use	0.526	0.531	0.094	.000
Perceived _usefulness -> Attitude toward _usage	0.609	0.608	0.094	.000
Perceived _usefulness -> Behavioral _intention to use	0.29	0.284	0.112	.009
Perceived ease of use -> Perceived _usefulness	0.877	0.88	0.03	.000

TABLE 6 Chatbot benefits perceived by students.

Easy to use	13.7%
It is embedded in the platform	11.0%
I can do the exercise many times until it works correctly	12.3%
Errors can be easily detected	35.6%
More effective learning	27.4%

and efficient responses to students when they inquire about errors found in the Java programming assignment significantly reduces wait times and improves the overall student experience. With the suggestions the chatbot gives to students, it creates individualized learning paths based on student needs and streamlines access to information, all contributing to a more positive, dynamic, and interactive learning environment.

This study highlights the observation that when students recognize the usefulness of chatbots in addressing their queries, providing feedback, and facilitating their learning journey, their overall attitude toward using these AI-driven tools becomes more positive. This aligns well with the case of Lola [35] and Dina [19], two chatbot applications used at the University of Murcia (Spain) and Dian Nuswantoro University (Indonesia), respectively, to manage enrollment-related queries. Students appreciate the convenience of quickly accessing information and receiving accurate and relevant responses to their queries [46]. Additionally, students expressed that one of the most important benefits was that the chatbot provided 24/7 support and was simple to use. By offering instant responses and finding that chatbots are easy to use, students perceive them as valuable tools, and their acceptance and satisfaction levels increase.

The findings also suggest that efforts to enhance the UI of chatbots can lead to increased acceptance because students who find chatbots easy to use were more likely to perceive them as valuable tools in their Java programming assignment. The experiment did not explicitly manipulate the UI of the chatbot as a controlled variable. However, the development process paid special attention to ease of use, intuitive navigation, and visual appeal when constructing and integrating the chatbot into the LMS. This emphasis on user-centric design aimed to create an enjoyable user experience for students interacting with the chatbot [30]. Well-designed UIs contribute to positive user experiences and can influence acceptance of technology [25]. In light of this, it is hypothesized that efforts to enhance the UI of chatbots, as reflected in our design approach, could potentially lead to increased acceptance among students. Although direct causal claims cannot be made based on the current study design, we must remember that Duolingo is praised for its effectiveness and user-friendly interface by incorporating game-like elements into the learning process [28], which can keep learners engaged and motivated [3], and more likely to achieve their learning goals. The learning activity in this study combined two interesting technologies, an IDE and a chatbot, and both increased students' motivation and participation. Chatbots can work well on their own, but when combined with another technology, better results are achieved [48].

9 | CONCLUSION AND FUTURE WORK

The goal of this study, based on the TAM model, was to illustrate a specific application of chatbots in education. The study demonstrated how a chatbot can be utilized for educational purposes, serving as a tutor to assist students in a course. It provided an opportunity to conduct a correlation analysis between the factors influencing the use and acceptance of this type of technology among university students.

These findings suggest that chatbots, when effectively integrated into the teaching process, can offer tangible benefits to teachers. Immediate feedback, quick access to information, increased engagement, personalized learning experiences, and time-saving aspects collectively support teachers in their endeavor to assist students with assignments.

The findings reveal a robust positive association between students' behavioral intention to use chatbots and their subsequent actual usage. When students clearly express their intent to integrate chatbots into their coursework, this inclination tends to translate into tangible adoption. This study explores important insights to improve strategies for increasing chatbot usage in educational settings. These strategies involve making the chatbots easier to use, highlighting their practical benefits, and fostering positive attitudes among students.

Teaching is a process of communication and interaction, a role that chatbots seamlessly handle due to their ability to understand natural language. Their true potential in education lies in fostering this interaction. The element of interactivity in online learning environments is crucial, and chatbots can provide it [47]. The effectiveness of a chatbot in improving the learning experience depends on how well it's designed and fits specific uses.

In the context of our study, the chatbot served not as a substitute for a teacher but rather as an assistant. Its primary function was addressing students' inquiries about the exercise, providing information, and guiding them to identify programming errors. This approach empowered students to make better decisions based on the chatbot's insights. The goal was to enhance the learning process by making the most of the chatbot's strengths within a defined scope, emphasizing ease of use, and aligning with the course's learning goals. This study introduced an initial version of the TAM model to evaluate students' acceptance of chatbots. However, in subsequent research phases, external variables may be incorporated into the model. This updated model is expected to provide valuable recommendations for optimizing the chatbot's configuration, aiming to achieve better results.

For the development of this second model, three external variables identified by Hong and Yu [26] in their literature review to determine the acceptance of Building Information Modeling (BIM) software may be incorporated. These variables include (a) utility and quality of the content, (b) flexibility, linked to external variables such as customization and instant connectivity, and (c) integration, related to the convenience of communication and information exchange in a collaborative and interactive environment. Figure 7 shows the extended model.

Future studies could also include a performance comparison involving two groups of students: one group using the chatbot and another not using it. Evaluating the impact on student performance is crucial in these types of studies, as it provides insights into the effectiveness of incorporating chatbots in educational settings. In the Latin American educational landscape, there are limited studies exploring this aspect. Our intention with this suggestion is to encourage further exploration and contribute to the existing body of knowledge, particularly in regions where such research might be less prevalent.

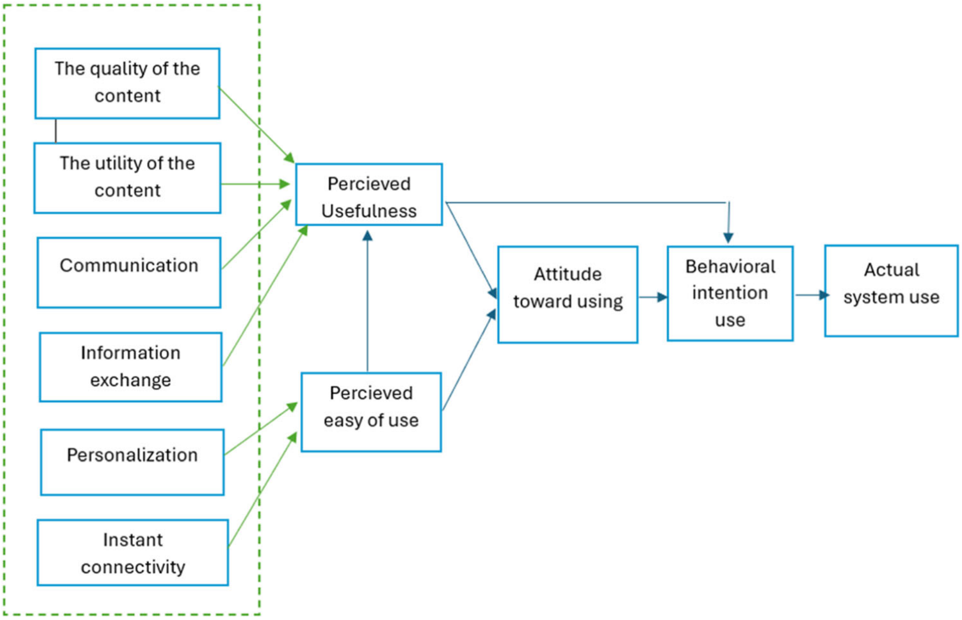

FIGURE 7 Extended Model.

Technology is ever-changing, and conversational agents such as ChatGPT, Gemini, and Copilot have emerged. However, the integration of these agents still has a long way to go. Their usage and applications seem limited only by instructors' creativity, as well as, by how they guide students to use this technology ethically and responsibly to enhance their learning.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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APPENDIX

Table A1

TABLE A1 Items used in the study instrument.

Construct	Number of questions	Item
PU	3 questions	(1) The JavaScript exercise was easier and faster to solve by using the assistance of a chatbot. (2) The chatbot helped me to understand, in a better way, the JavaScript programming concepts. (3) In general, do you consider that using a chatbot assistant within the course was useful to learn?
PEU	2 questions	(1) Learning to use the Chatbot was easy for me. (2) It was easy to get good materials from the chatbot.
ATU	4 questions	(1) This technology (AI chatbots) makes learning more comfortable, efficient, and secure. (2) I consider that the assistance of a chatbot to solve a JavaScript programming exercise was useful to reinforce the concepts learned during lectures. (3) I am satisfied with this type of exercise. (4) I consider that using a chatbot to solve the exercise is a good idea.
BIU	3 questions	(1) Teachers should consider doing more assignments like this one. (2) I think the chatbot was useful for me as a student. (3) After completing the exercise, I would like to continue using a chatbot to help me with my assignments.
Additional Variables		(1) Age. (2) Gender. (3) Technology Usage level. (4) Knowledge of chatbots.

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Mónica De La Roca is a computer sciences engineer and has an MSc in Data Base from the University of Francisco Marroquín (Guatemala). She has an MSc in e-learning from the University of Carlos III of Madrid and is a PhD candidate from the University of Alcalá (Spain). Currently, she coordinates the instructional design division at the Research and Development Department (GES) at Galileo University (Guatemala). As a MOOCs specialist, she is part of the GalileoX team in edX. She is now a professor of courses related to digital education and technology at the Institute Von Neumann at Galileo University. Her research interests include digital transformation, MOOCs, learning methodologies, digital competencies, and so forth.

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